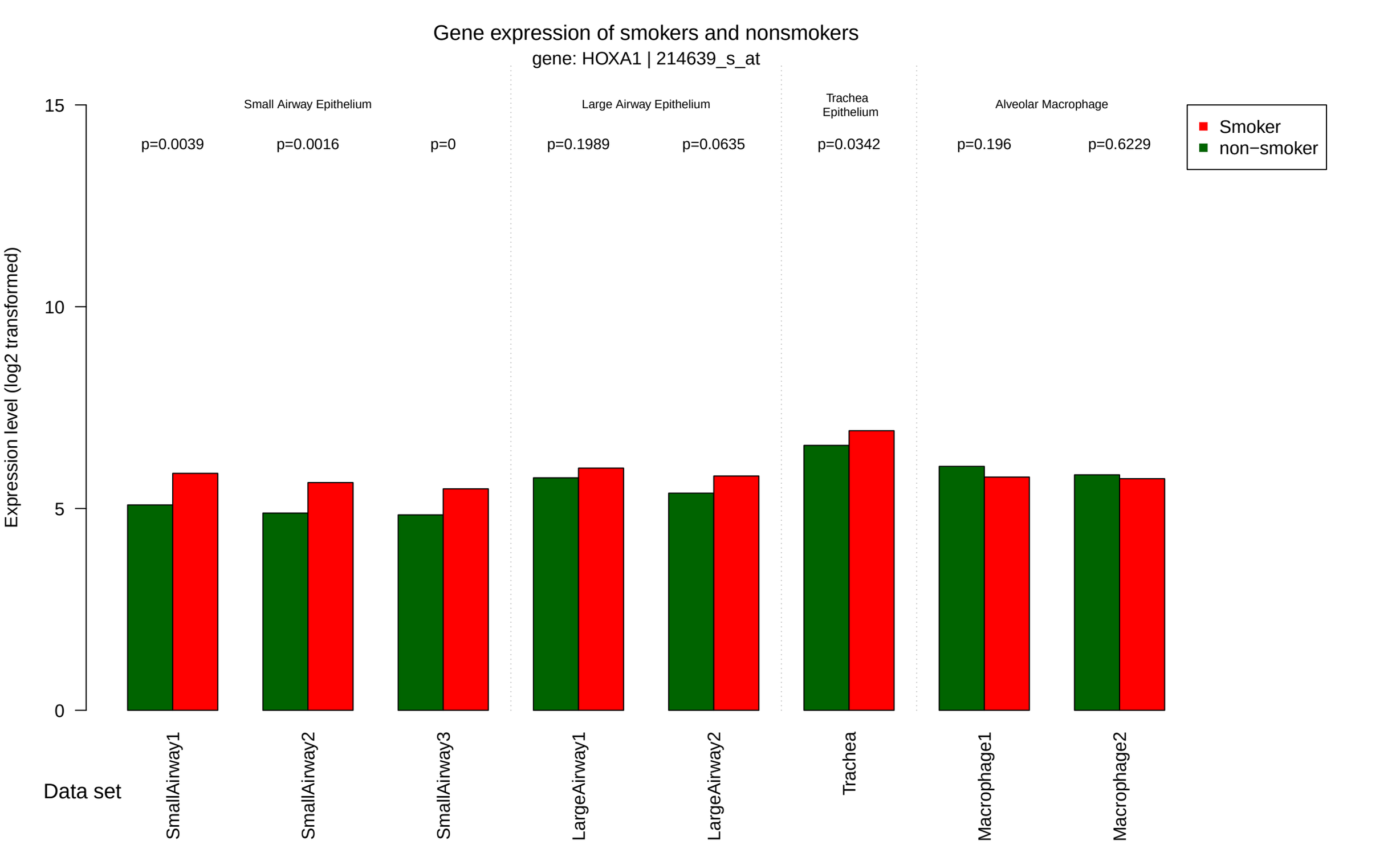
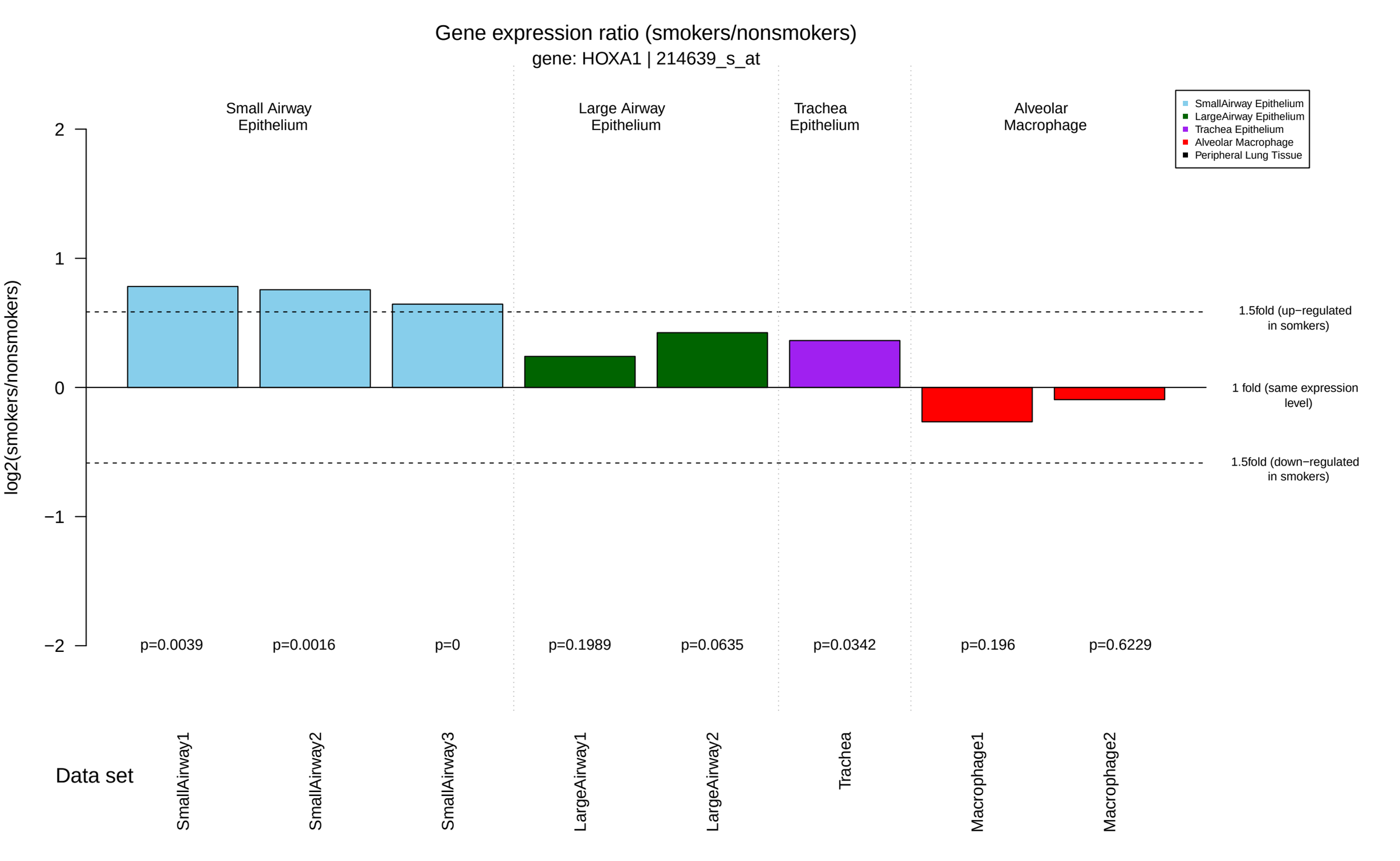
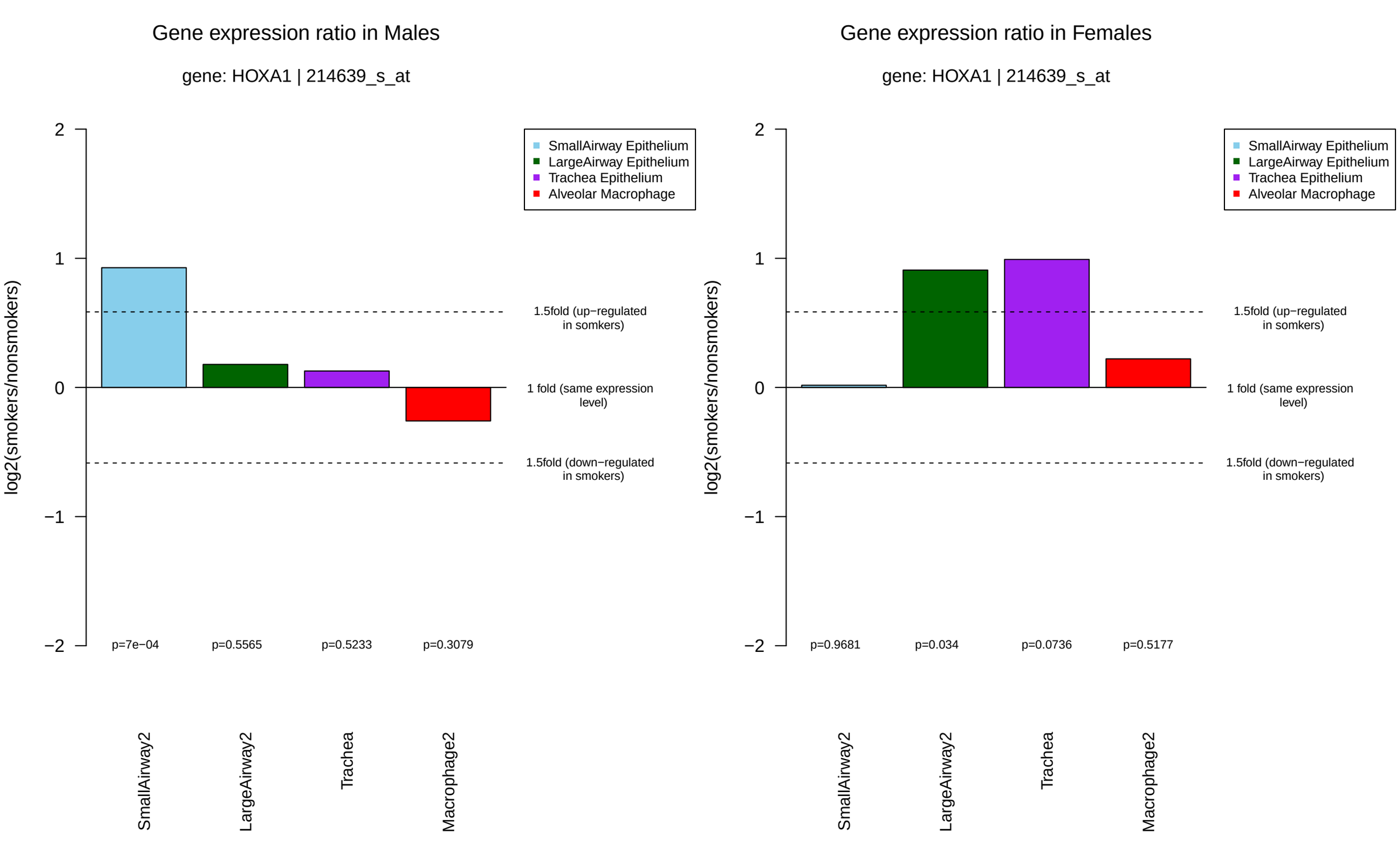
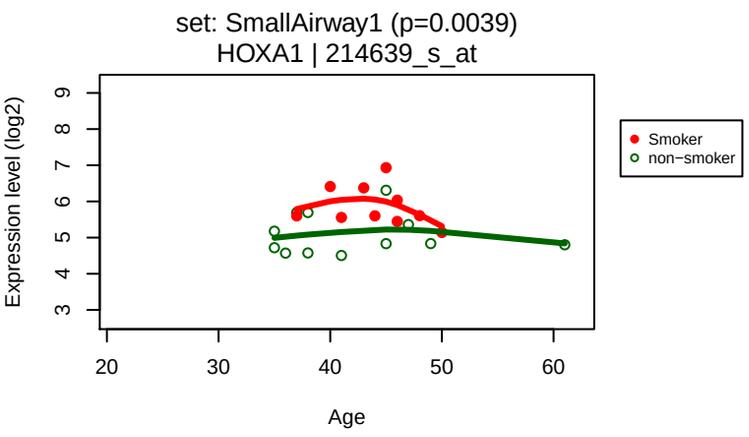
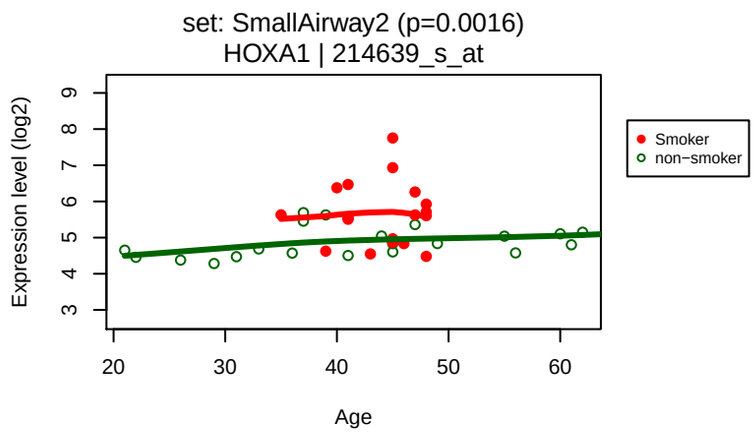
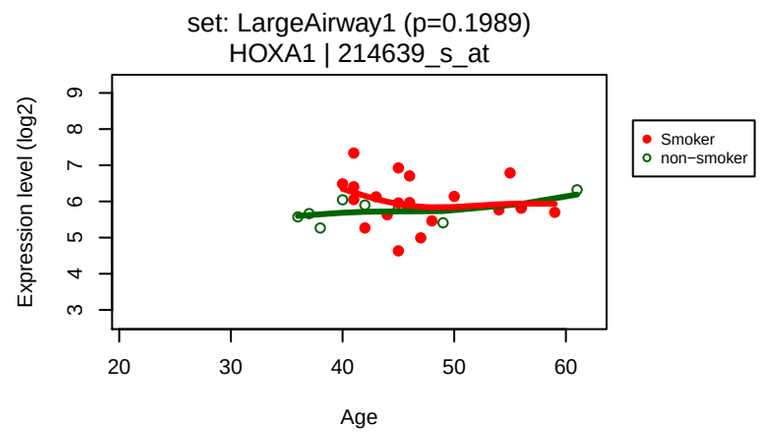

Gene expression of smokers and nonsmokers
gene: HOXA1 | 214639_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0039
p=0.0016
p=0
p=0.1989
p=0.0635
p=0.0342
p=0.196
p=0.6229
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

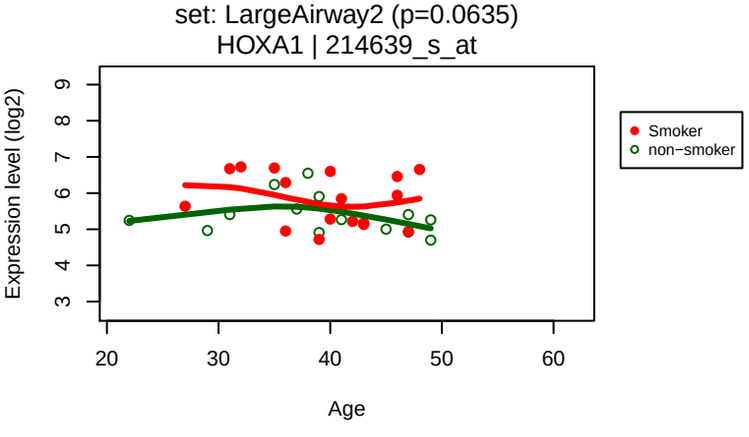

Gene expression ratio (smokers/nonsmokers)
gene: HOXA1 | 214639_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0039
p=0.0016
p=0
p=0.1989
p=0.0635
p=0.0342
p=0.196
p=0.6229
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

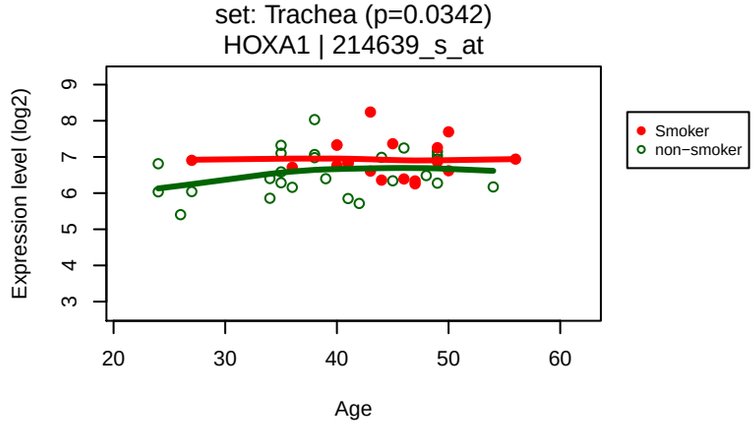

Gene expression ratio in Males
gene: HOXA1 | 214639_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=7e−04
p=0.5565
p=0.5233
p=0.3079
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXA1 | 214639_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9681
p=0.034
p=0.0736
p=0.5177
SmallAirway2
LargeAirway2
Trachea
Macrophage2

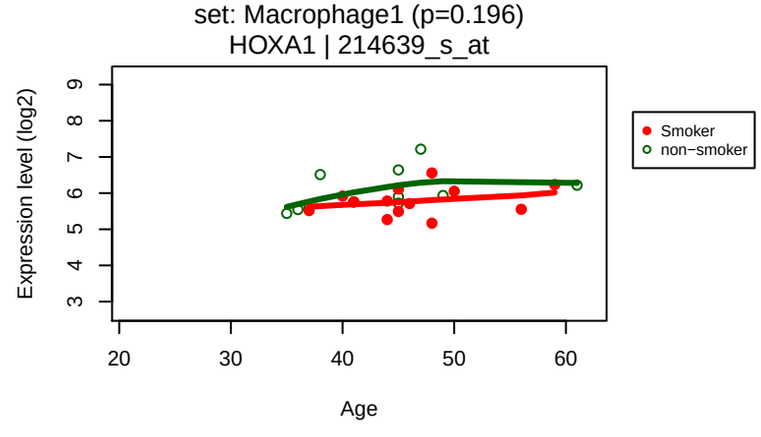

set: SmallAirway1 (p=0.0039)
HOXA1 | 214639_s_at
Expression level (log2)
Age
Smoker
non−smoker

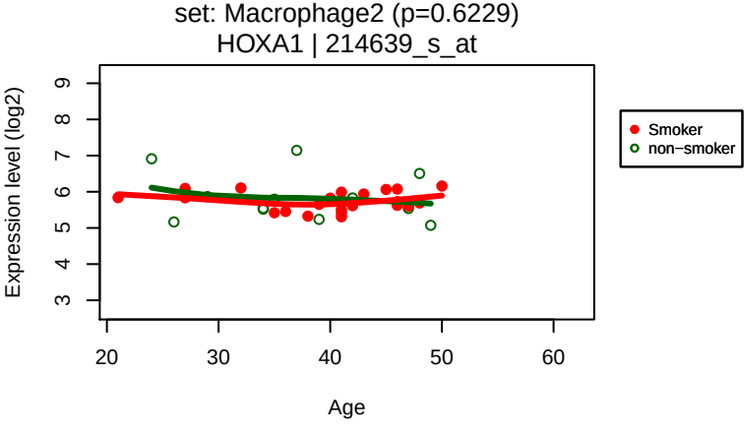

set: SmallAirway2 (p=0.0016)
HOXA1 | 214639_s_at
Expression level (log2)
Age
Smoker
non−smoker

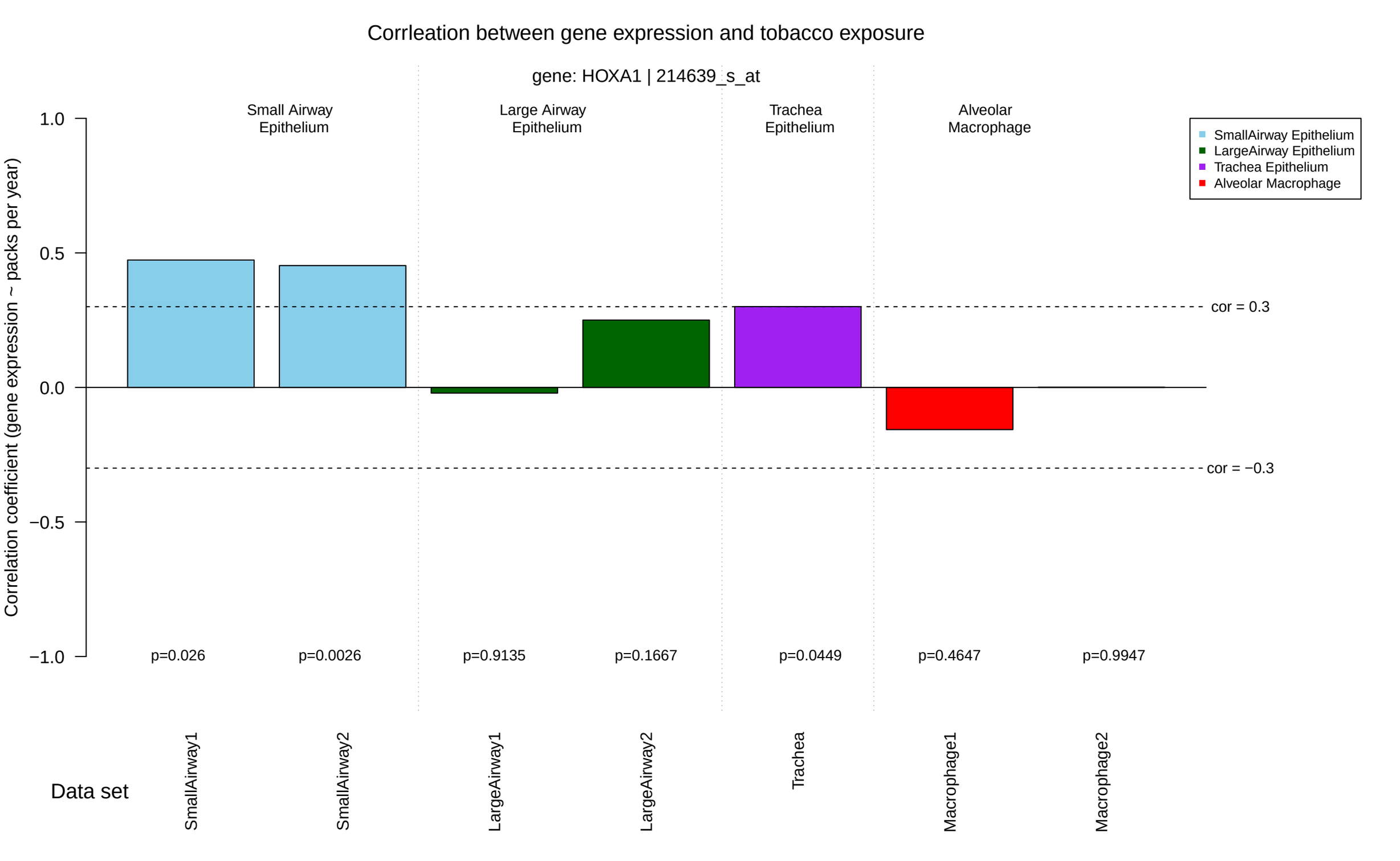

set: LargeAirway1 (p=0.1989)
HOXA1 | 214639_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0635)
HOXA1 | 214639_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0342)
HOXA1 | 214639_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.196)
HOXA1 | 214639_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6229)
HOXA1 | 214639_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXA1 | 214639_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.026
p=0.0026
p=0.9135
p=0.1667
p=0.0449
p=0.4647
p=0.9947
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2